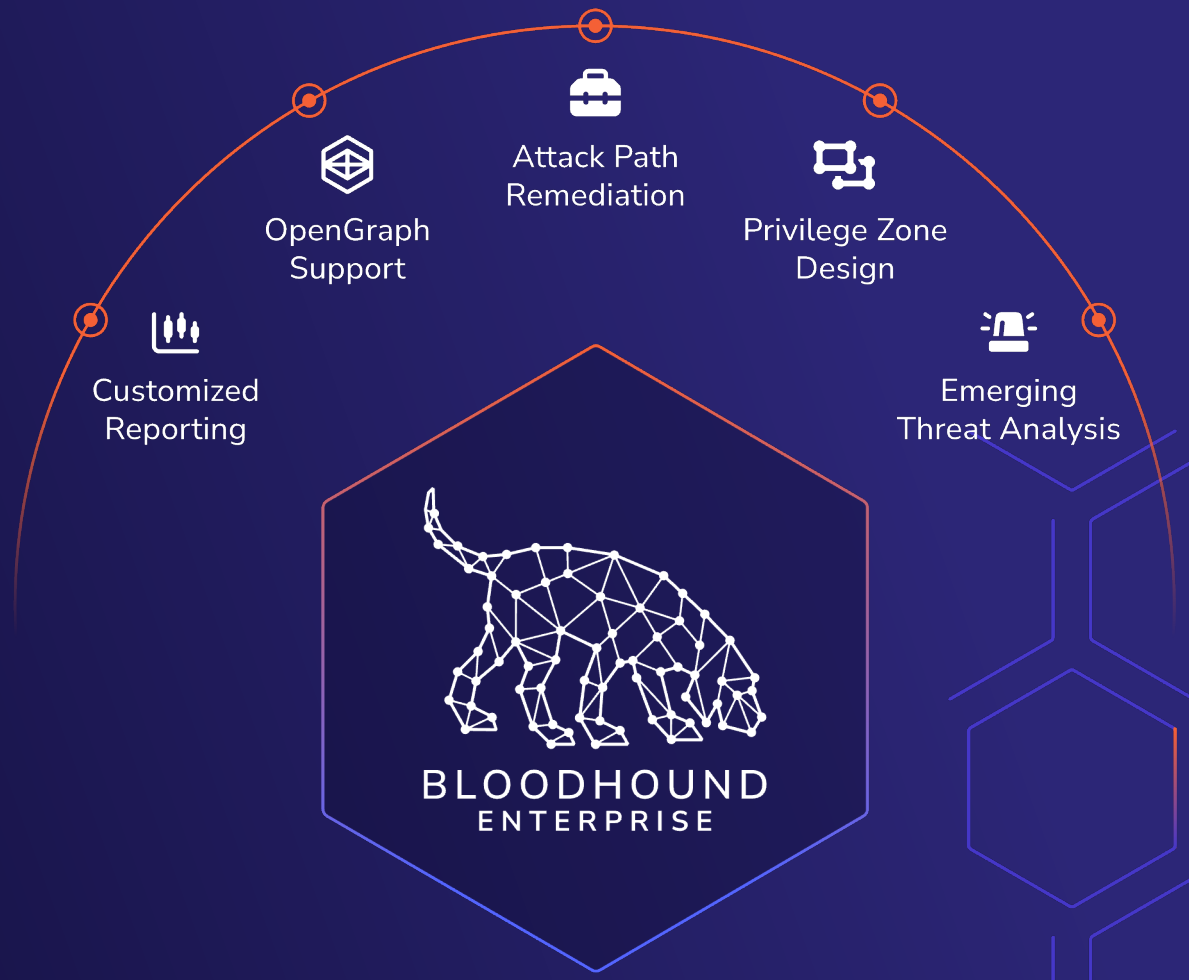
SPECTEROPS

BloodHound Scentry

Accelerate Your Identity APM Practice



Robby Winchester

Robby Winchester

Chief Global Professional Services Officer

- SpecterOps Co-founder
- Experienced threat hunter and penetration tester serving private-sector and government clients
- Developed and led penetration testing, physical security, and breach assessments across multiple organizations
- Former U.S. Air Force Information Aggressor, conducting full-scope network and physical red team assessments

© Specter Ops, Inc.

Built by attackers.
Trusted by defenders.

Unmatched offensive security services

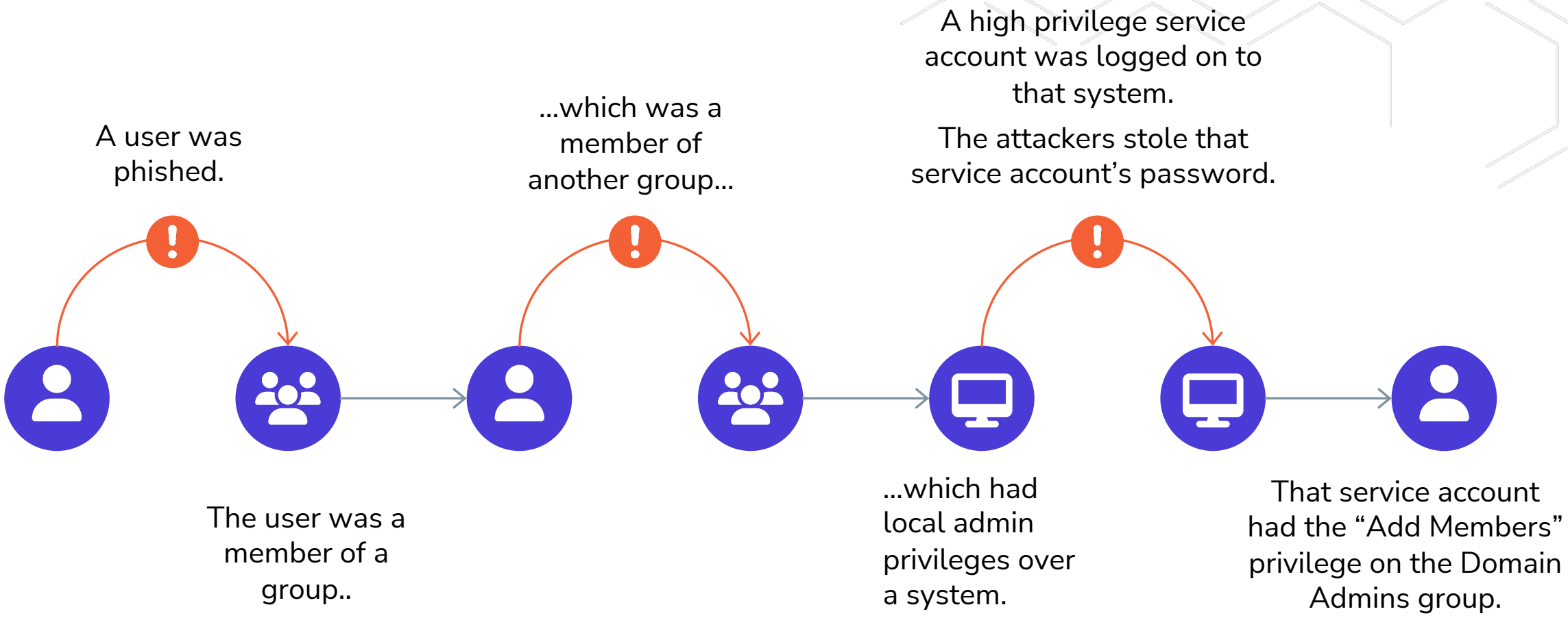
Innovative research, tradecraft & training

Industry defining open-source projects

Enterprise-grade attack path platform

© Specter Ops, Inc.

How attackers look at identity relationships



© Specter Ops, Inc.

The Birth of BloodHound

Defenders think in lists; Attackers think in graphs

- Released in 2016, BloodHound enabled Red Team Operators to quickly find fastest path from initial foothold to Tier 0 asset
- Improves operational efficiency by removing hours of tedious, manual effort mapping identity relationships
- Provided new framework for novel attack graph research and identification of new identity attack primitives
- Has become essential solution for penetration tests and red team activities

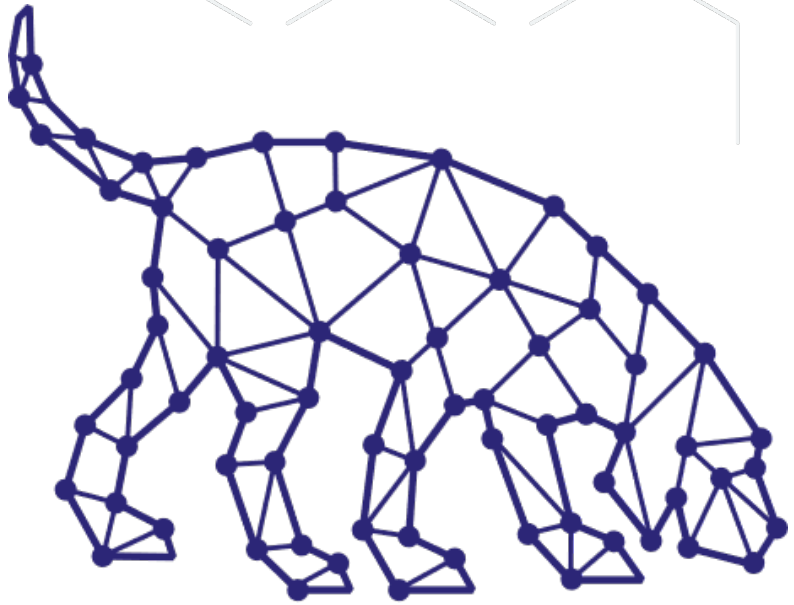
BLOODHOUND
COMMUNITY EDITION

© Specter Ops, Inc.

The Evolution of a BloodHound

To Stop an Attacker, You Must Think Like an Attacker

- Provides attack path visibility scaled to all identities, and critical assets in an environment
- Measures exposure of all attack paths to provide prioritize roadmap for removal
- Safe, tested remediation guidance targeted at reducing the most attack path exposure with each action
- Enterprise integrations across existing Identity, Security, and IT



BLOODHOUND
ENTERPRISE

© Specter Ops, Inc.

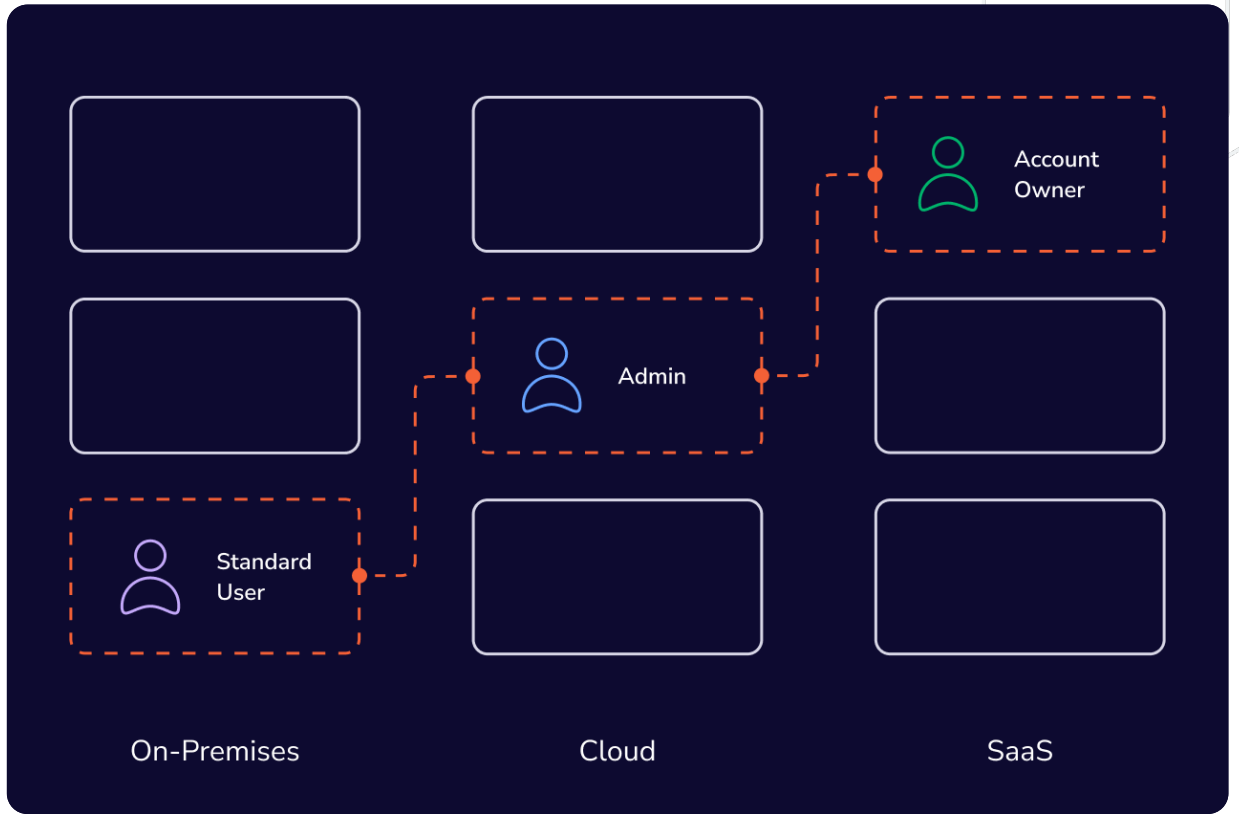
Attack Paths Don’t Stop at Admin Access

Neither Does BloodHound Enterprise

Attackers aren’t only focused on the control plane – they want to use their access to maximize the reach and impact of their operation. Expand BloodHound Enterprise attack path management to protect your critical assets

Privilege Zones Protect:

- Regulatory enclaves
- Sensitive customer data
- Code repositories
- Vital business applications
- and more



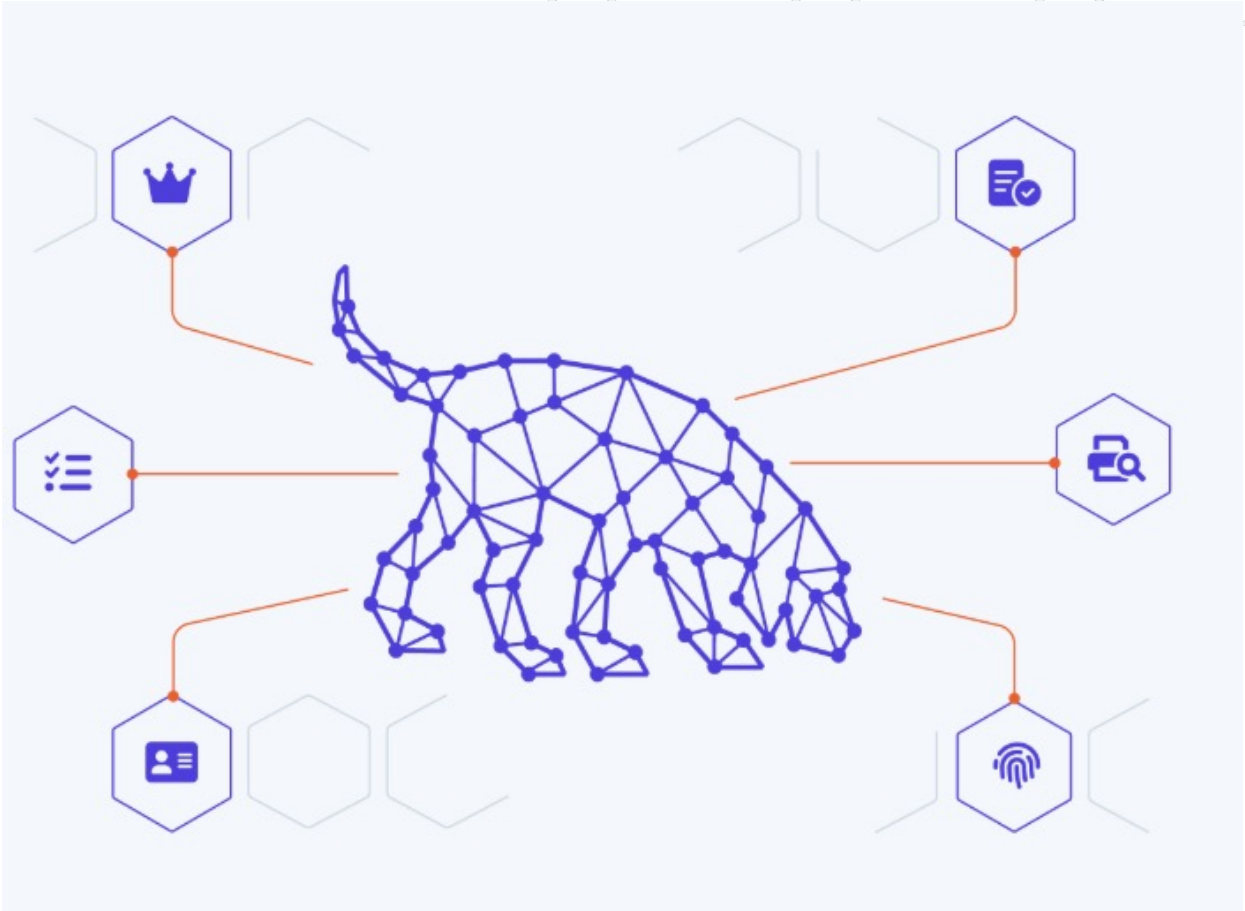
SPECTEROPS

© Specter Ops, Inc.

OpenGraph takes BloodHound Even Further

Take Off the Leash

- Harness BloodHound's power to uncover hidden attack paths throughout your technology stack
- Visualize custom attack paths across cloud services, SaaS applications, and even internal custom tools
- Continuously expanding OpenGraph Extension Library – with 30+ new extensions ...so far



© Specter Ops, Inc.

APM Maturity Model

Six levels of capability, ranging from absence of practice to preventive privilege design

0 Nonexistent

- Attack paths unknown
- Response occurs after incidents/ pen tests
- No visibility, ownership, or tracking

1 Initial

- Remediation is manual and responsive
- Informal ownership
- No measurement or prioritization

2 Managed

- Documented processes
- Efforts remain siloed and qualitative
- Risk prioritization is subjective

3 Defined

- Enterprise tooling with risk quantification
- Continuous risk-based remediation
- Alignment between Identity and Security

4 Quantitatively Managed

- Identity APM insights embedded in IAM, IGA, and IR
- Common KPIs and shared risk model
- Identity APM data is the source of truth for identity privilege exposure

5 Optimizing

- Attack paths avoided by design
- APM data actively shapes identity decisions
- Preventive controls are enforced via policy

© Specter Ops, Inc.

Real business and operational value

> The complexity of identity security urgently needs solving at the enterprise level and I believe the Attack Path Management approach from SpecterOps is that solution."

Kevin Mandia
Chairman of the Board, SpecterOps
Co-Founder & Partner, Ballistic Ventures
Founder of Mandiant Security

FORTUNE 100 FOOD & BEV

2 million identities

54% reduction

in exposure in 6 months

FORTUNE 100 BANK

50 domains

100% elimination

of Tier 0 attack paths in 6 months

FORTUNE 100 HEALTHCARE

4 million identities

90% reduction

in attack paths in 18 months

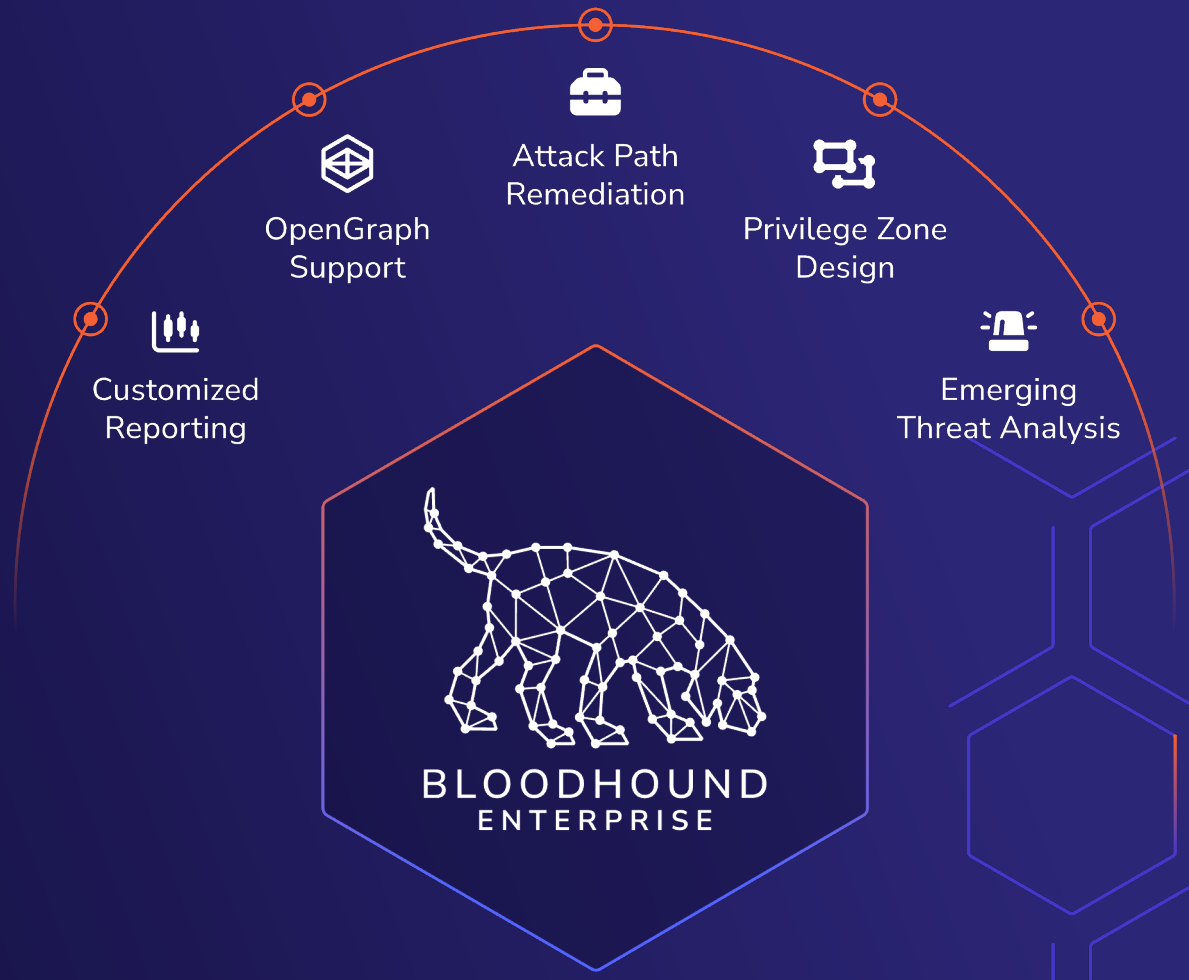
* Legal Disclaimer (CELA): The quote above reflects my personal opinion and does not represent the view of Microsoft or its affiliates.

© Specter Ops, Inc.

SPECTEROPS

BloodHound Scentry

Accelerate Your Identity APM Practice



Accelerate your practice and impact

- Remediate attack paths with full context of your operations
- Protect the assets most important to your business
- Stay ahead of emerging threats that affect your industry
- Expand your practice to new assets and environments
- Prove results to the business

Expert Analysis

Monthly analysis of attack paths and trends

Monthly review session of findings and actions

Emerging threat analysis

Privilege Zone Design

Critical asset discovery and prioritization

Configuration of Privilege Zones

Ongoing governance and certification

BloodHound OpenGraph Support

Configuration and deployment

New platform expansion and coverage

Custom OpenGraph dev and support

BloodHound Scentry

Attack Path Remediation

Advanced analytics and impact assessment

Tailored remediation guidance

High value target remediation identification

Customized Reporting

Executive APM reports

MTTR tracking

Path Remediation metrics

SPECTEROPS

© Specter Ops, Inc.

Advance Attack Path Management

BloodHound Scentry gets you there faster

0 → **5**

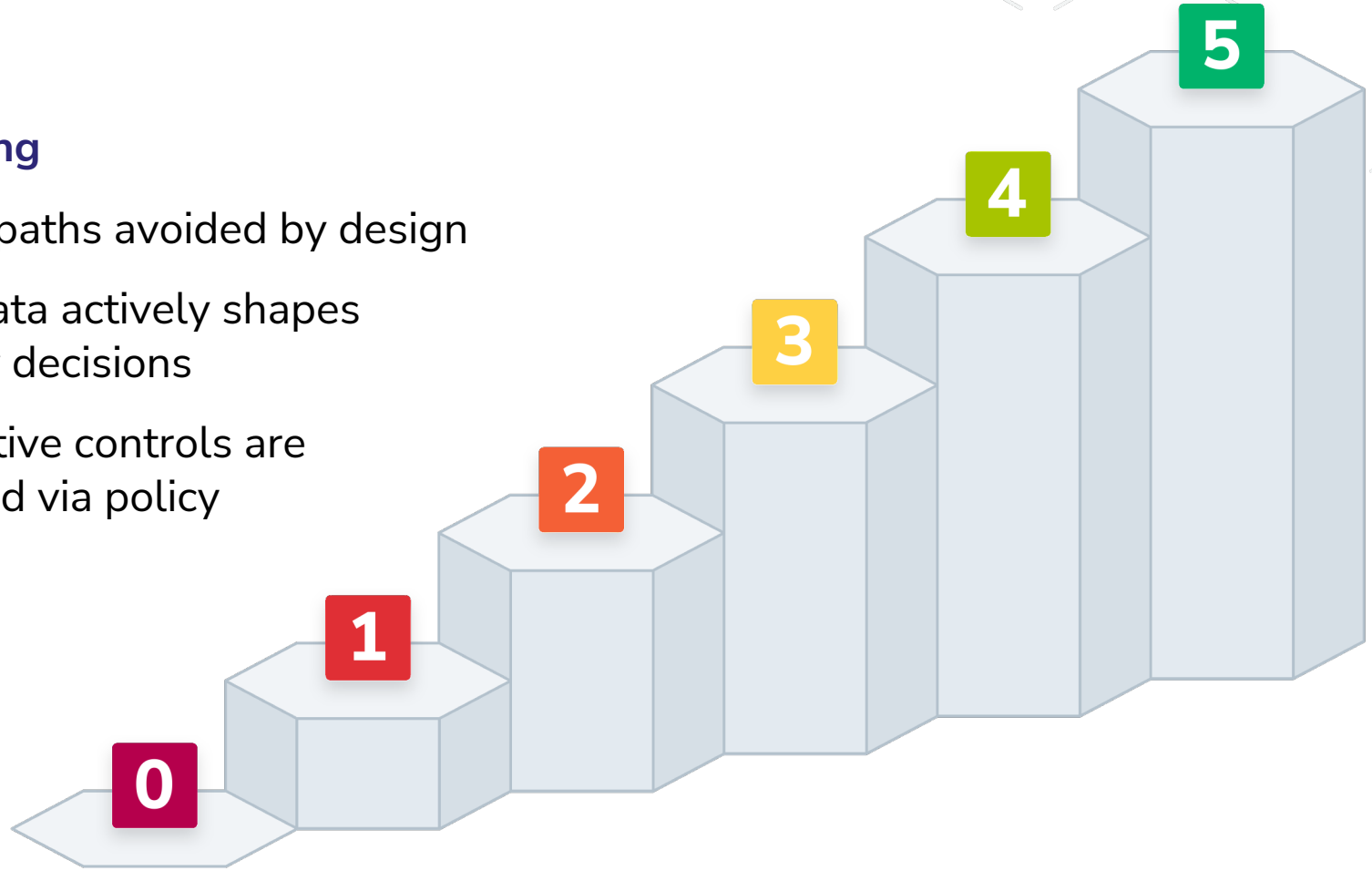
Nonexistent

- Attack paths unknown
- Response occurs after incidents/pen tests
- No visibility, ownership, or tracking

Optimizing

- Attack paths avoided by design
- APM data actively shapes identity decisions
- Preventive controls are enforced via policy

Ask your account team for the full APM Maturity Model



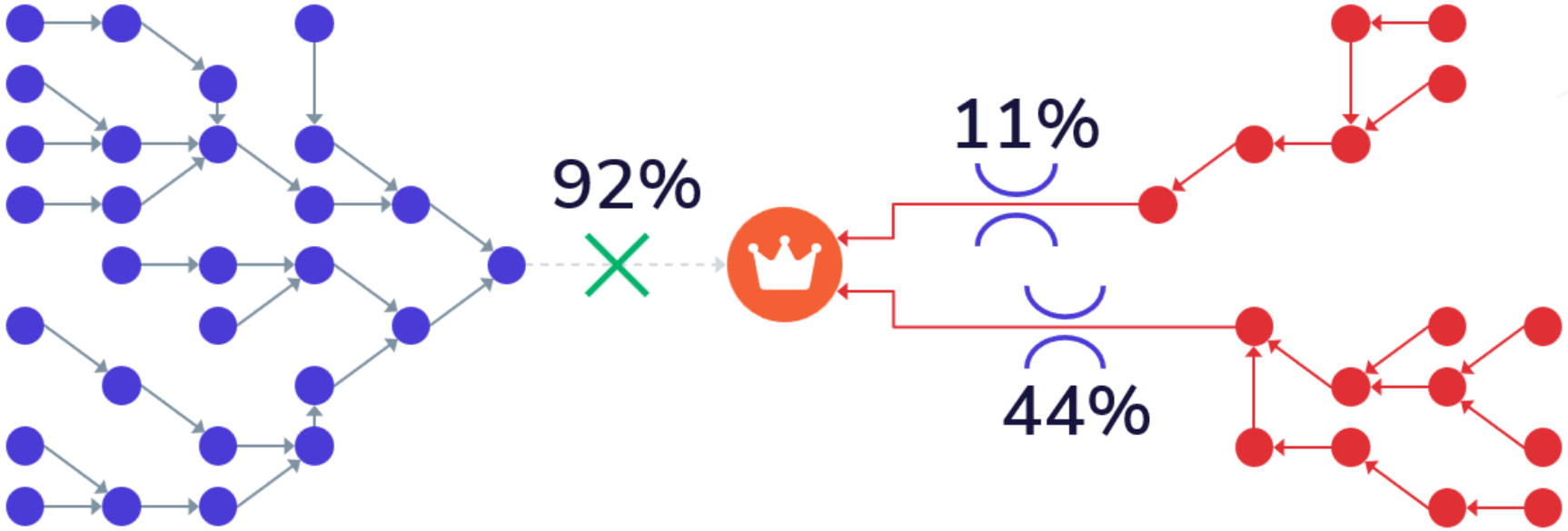
© Specter Ops, Inc.

Remediate with full context

“What do I fix next?” guidance

Tailored remediation recommendations and impact assessment based on analysis of your specific graph

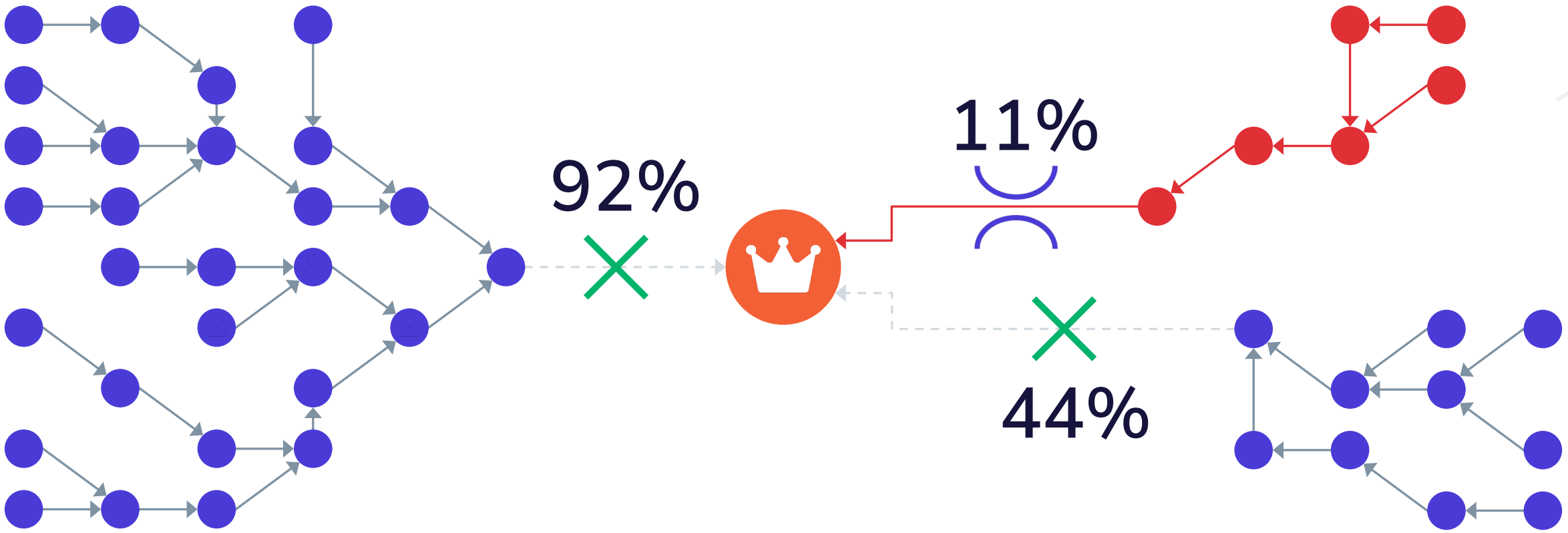
- Expert-led, step-by-step remediation playbooks
- Custom analysis specific to your environment
- Expert-supported explanations for each recommended action



© Specter Ops, Inc.

Remediate with full context

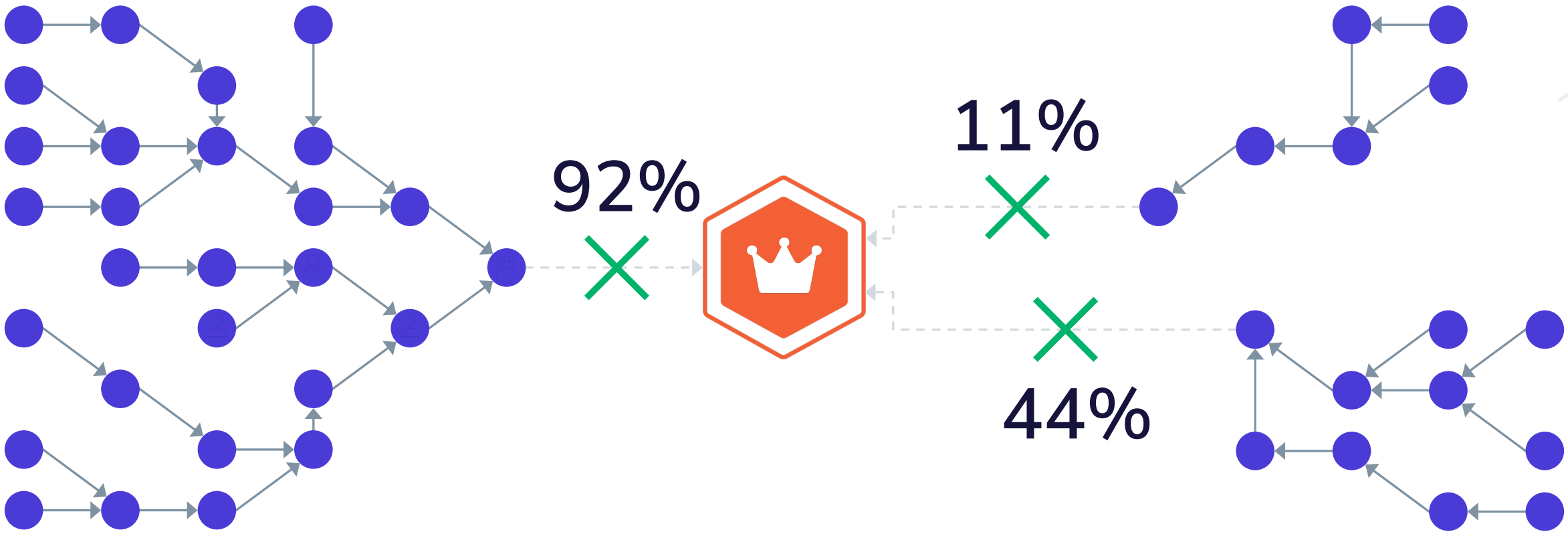
“What do I fix next?” guidance



© Specter Ops, Inc.

Remediate with full context

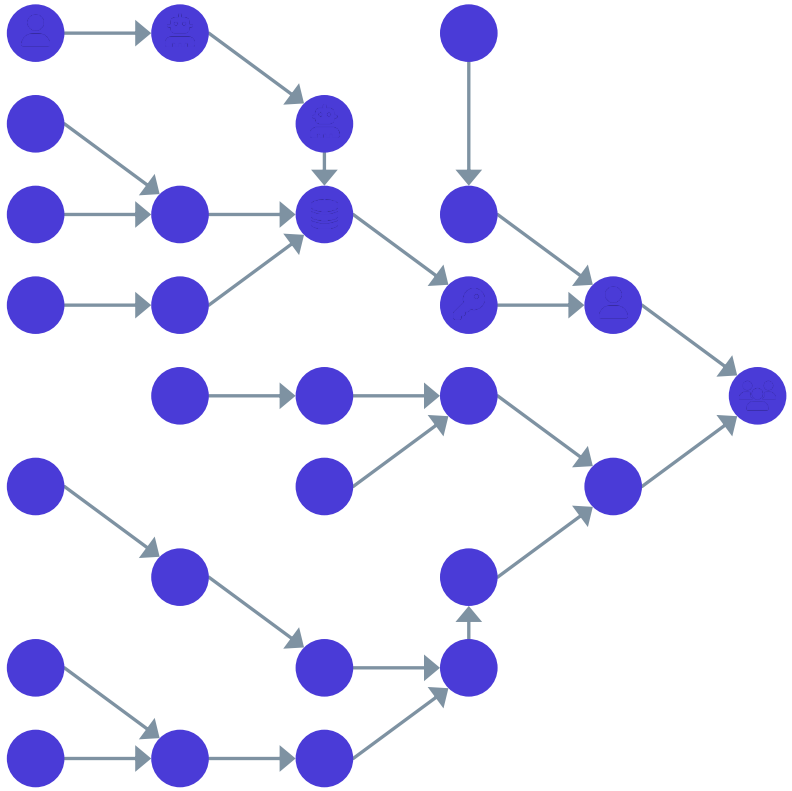
“What do I fix next?” guidance

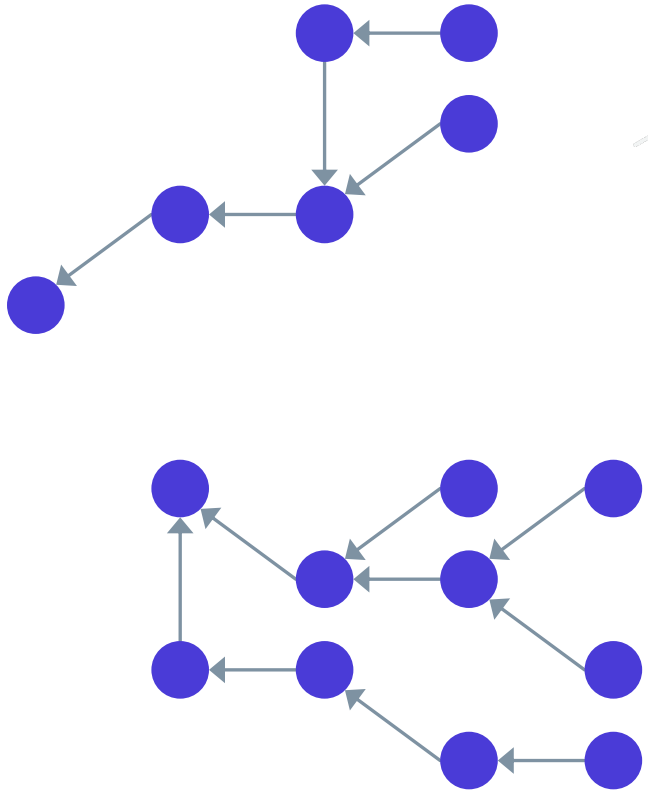


© Specter Ops, Inc.

Remediate with full context

“What do I fix next?” guidance





© Specter Ops, Inc.

Design Privilege Zones for critical assets

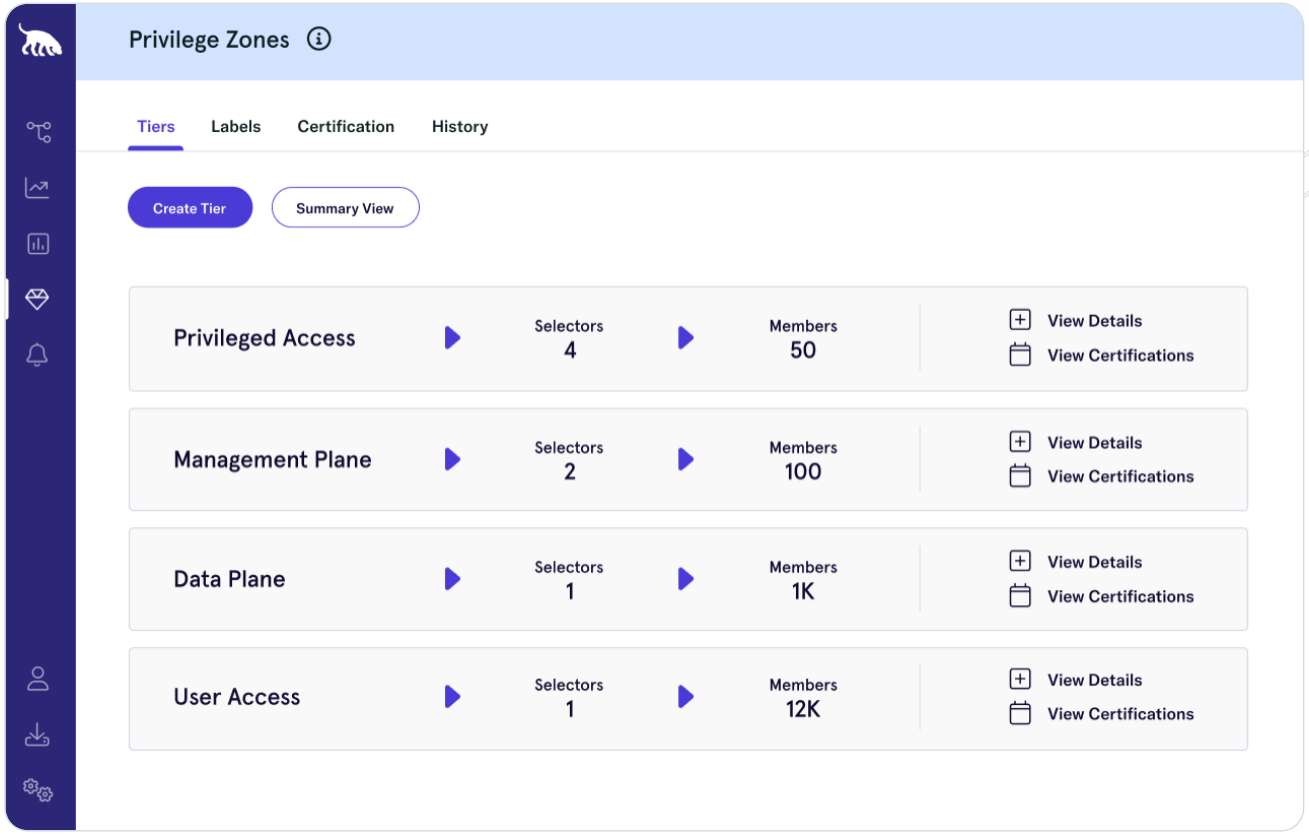
Identify, prioritize, and protect what's important to your business

Includes

- Interview with key stakeholders
- Identification of critical assets for protection
- Guidance for implementing Privilege Zones

Deliverables

- Privilege Zone design document
- Library of selectors and labels
- PZ certification and governance workflows
- OpenGraph integration notes (as applicable)

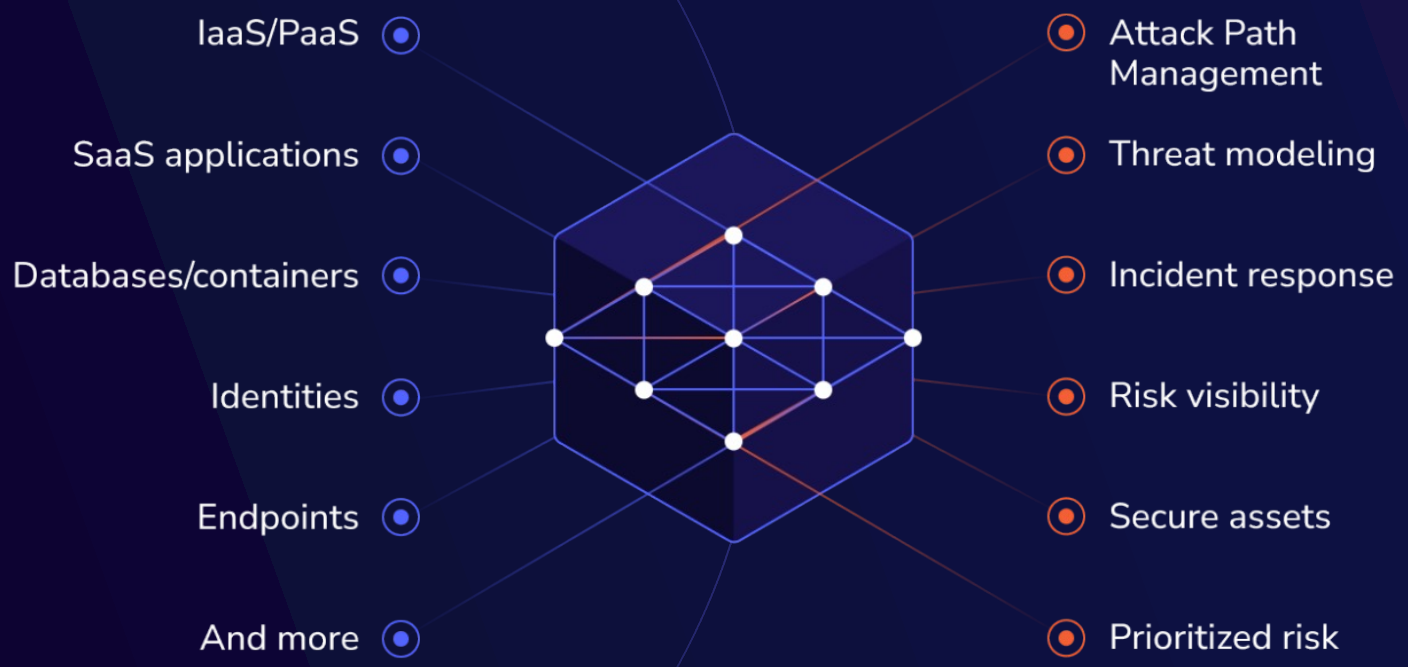


© Specter Ops, Inc.

Expand coverage with BloodHound OpenGraph

BloodHound OpenGraph extends APM to identify threats specific to your unique environment

- Model identity risk across disparate systems and applications
- Visualize custom attack paths within cloud services, SaaS applications, and even internal custom tools
- Harness BloodHound's robust search and Cypher query capabilities to uncover hidden attack vectors throughout your technology stack



SPECTEROPS

© Specter Ops, Inc.

One Year, 30+ OpenGraph Extensions

Supporting Security Research Into the Platforms You Rely On

- Ansible
- Active Directory (AD)
- Amazon Web Service (AWS)
- Atlassian
- Cisco Duo Security
- Credentials
- CyberArk
- Google Cloud Platform (GCP)
- Jamf
- MSSQL
- Network
- Entra ID
- FreeIPA
- GitHub
- Ping
- runZero
- Snowflake
- System Center
- vCenter
- Windows

This is just the beginning of OpenGraph

SPECTEROPS

© Specter Ops, Inc.

Gain Expert Analysis

Identify new threats important to your industry

- Advanced tradecraft applied to find additional risk
- Analysis of findings that are not attack paths but warrant investigation
- Identification of technologies that would benefit from OpenGraph extensions
- Monitoring of relevant channels for emerging TTPs that may impact your environment
- Reports scoped to each emerging threat delivered for review and discussion

© Specter Ops, Inc.

Expert Analysis Story

Challenging Even the Most Ambitious Paradigms

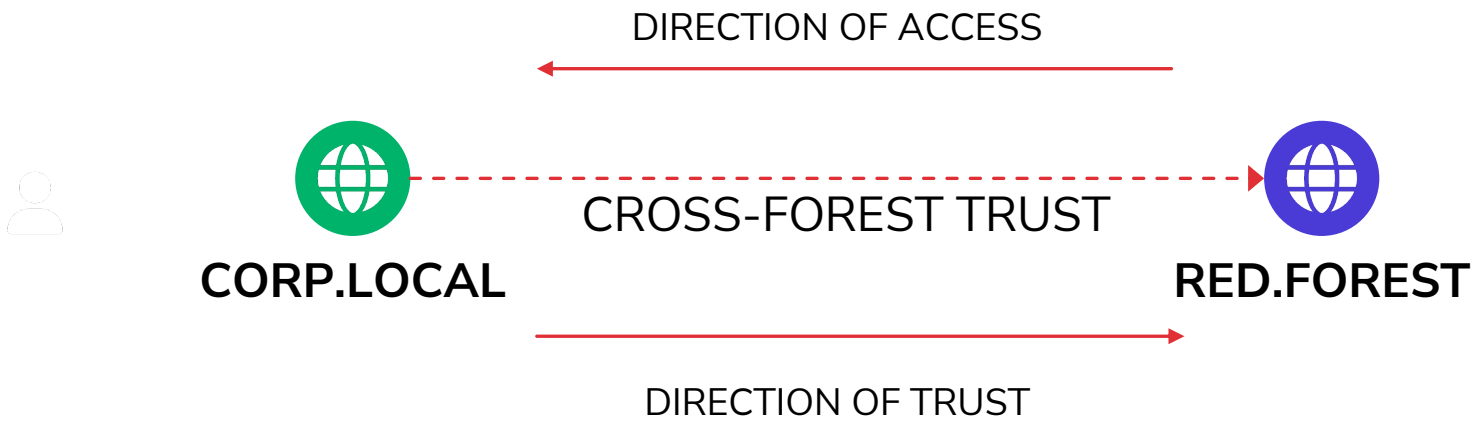
- Microsoft's Enhanced Security Admin Environment is a legacy security model for Active Directory
- AKA "Red Forest"
- Complex to deploy, difficult to maintain
- Now deprecated in favor of the Enterprise Access Model

© Specter Ops, Inc.

Expert Analysis Story

Challenging Even the Most Ambitious Paradigms

- RED.FOREST authenticates to CORP.LOCAL for administrative tasks
- Direction of access is the opposite of the direction of trust
- CORP.LOCAL should not be able to authenticate to RED.FOREST

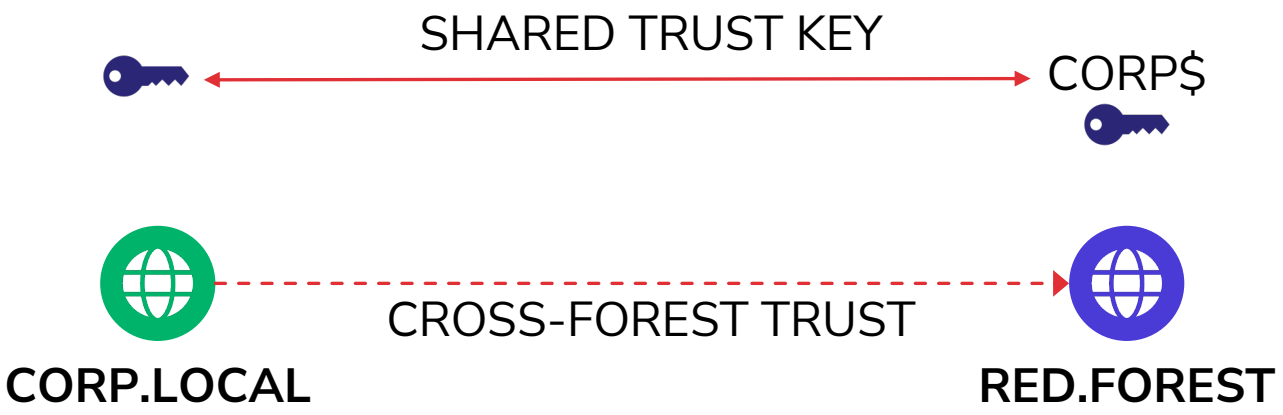


© Specter Ops, Inc.

Expert Analysis Story

Challenging Even the Most Ambitious Paradigms

- RED.FOREST contains CORP$ trust account
- Trust account key is shared with trusted domain object (TDO) in CORP.LOCAL

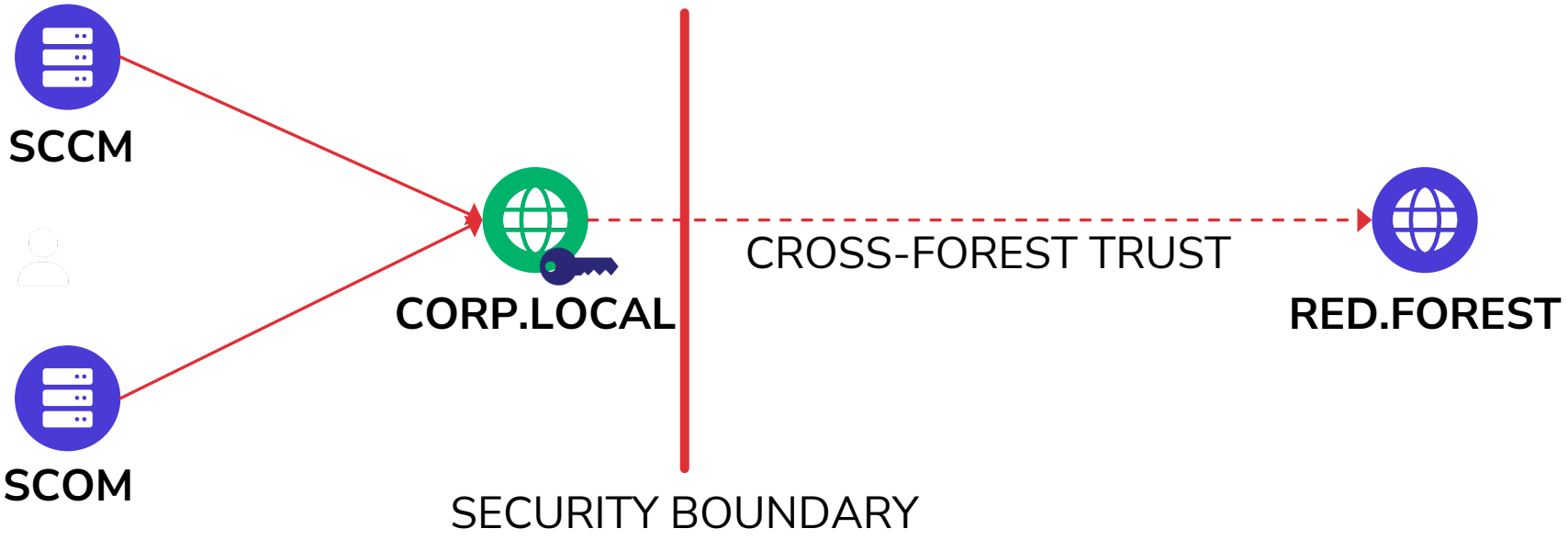


© Specter Ops, Inc.

Expert Analysis Story

Challenging Even the Most Ambitious Paradigms

- Compromise CORP.LOCAL through attack path to dump the TDO key

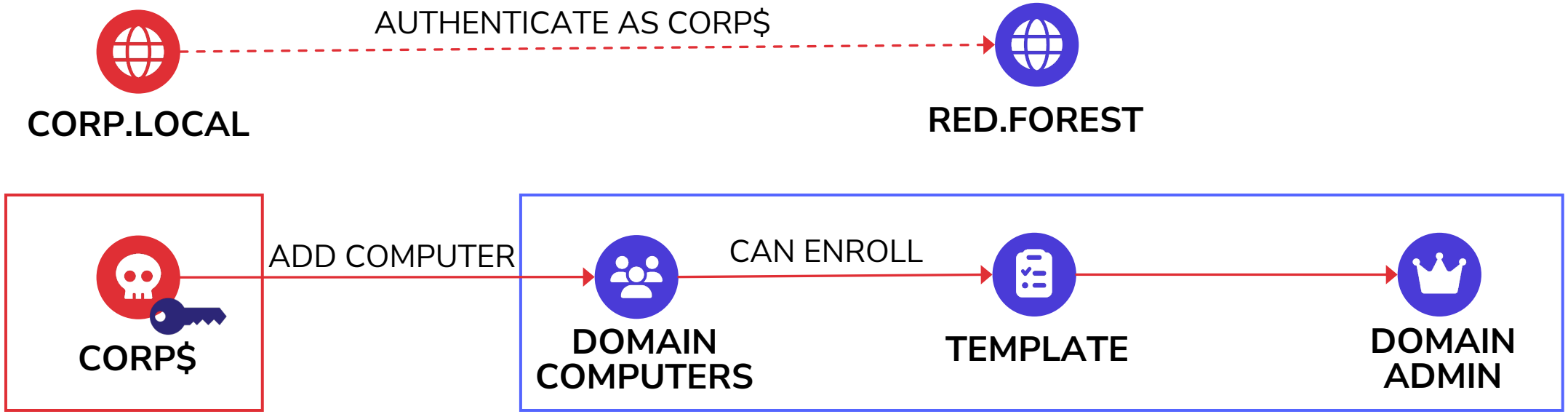


© Specter Ops, Inc.

Expert Analysis Story

Challenging Even the Most Ambitious Paradigms

- Authenticate as CORP$ to RED.FOREST
- ms-DS-MachineAccountQuota = 10
- ADCS ESC1 on DOMAIN COMPUTERS@RED.FOREST

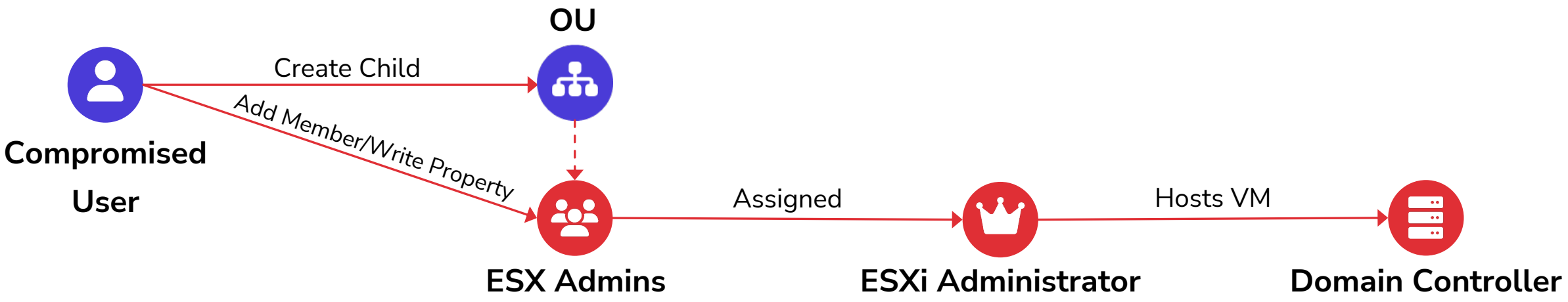


© Specter Ops, Inc.

Get Ahead of Emerging Threats

ESX Admins (CVE-2024-37085)

- Domain-joined ESXi hypervisors automatically grant full admin to members of "ESX Admins" group
- ESX Admins does not exist by default
- Any user with group-creation privileges can create it, add themselves, and escalate privileges to ESXi administrator
- OR any user with "WriteProperty" on *any* group can rename it



© Specter Ops, Inc.

Custom Reporting

Proving the impact of your attack path management practice

Scentry tracks the progress of your Identity Attack Path Management (APM) practice, connecting attack path remediation and privilege zone adoption to measurable improvements in identity security posture, operational risk reduction, and business outcomes.

Types of reports:

- Quarterly reporting packages and bespoke reporting based on your requirements.
- Executive summary – *What did the team fix over a period and what were the changes?*
- Identity Security Posture Improvements
- Compliance mapping
- MTTR Data

© Specter Ops, Inc.

Tailored specifically to your needs

Attack Path Remediation

- Advanced analytics and impact assessment
- Tailored remediation guidance
- High value target remediation identification

Privilege Zone Design

- Critical asset discovery and prioritization
- Configuration of Privilege Zones
- Ongoing governance and certification

Expert Analysis

- Monthly analysis of attack paths and trends
- Monthly review session of findings and actions
- Emerging threat analysis

BloodHound OpenGraph Support

- Configuration and deployment
- New platform expansion and coverage
- Custom OpenGraph dev and support

Customized Reporting

- Executive APM reports
- MTTR tracking
- Path Remediation metrics

Scentry is powered by experts in adversary tradecraft and attack path management

© Specter Ops, Inc.

Questions?

© Specter Ops, Inc.

SPECTEROPS
Creators of BloodHound

Thank you

specterops.io

© Specter Ops, Inc.